Yushi Huang

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Education

The Hong Kong University of Science and Technology — Hong Kong SAR
Ph.D. in Electronic and Computer Engineering; advisor: Prof. Jun Zhang — *Feb. 2025 – Present*

Beihang University — Beijing, China
B.Eng. in Computer Science and Technology — *Sep. 2020 – Jun. 2024*

Research Interests

My research interests focus on model efficiency and reinforcement learning (RL). I currently work on agentic RL for (M)LLMs; my prior work spans RL for image and video generation and inference acceleration for large foundation models (e.g., quantization, computation skipping, and efficient attention).

Selected Publications

[*] equal contribution; [†] corresponding author. Full list

- **MeanFlowNFT: Bringing Forward-Process RL to Average-Velocity Generators**
  Yushi Huang[*], Xiangxin Zhou[*†], Jun Zhang, Liefeng Bo, Tianyu Pang[†]
  Preprint, 2026.
- **MoDES: Accelerating Mixture-of-Experts Multimodal Large Language Models via Dynamic Expert Skipping**
  Yushi Huang, Zining Wang, Zhihang Yuan[†], Yifu Ding, Ruihao Gong, Jinyang Guo, Xianglong Liu, Jun Zhang[†]
  CVPR 2026.
- **LinVideo: A Post-Training Framework towards $\mathcal{O}(n)$ Attention in Efficient Video Generation**
  Yushi Huang, Xingtong Ge, Ruihao Gong[†], Chengtao Lv, Jun Zhang[†]
  CVPR 2026. ***Highlight.***
- **QVGen: Pushing the Limit of Quantized Video Generative Models**
  Yushi Huang, Ruihao Gong[†], Jing Liu, Yifu Ding, Chengtao Lv, Haotong Qin, Jun Zhang[†]
  ICLR 2026. ***Top 1.3%.***
- **Reinforcing Few-Step Generators via Reward-Tilted Distribution Matching**
  Yushi Huang[*], Xiangxin Zhou[*], Ruoyu Wang[*], Chi Zhang, Jun Zhang, Tianyu Pang[†]
  Preprint, 2026.
- **HarmoniCa: Harmonizing Training and Inference for Better Feature Caching in Diffusion Transformer Acceleration**
  Yushi Huang[*], Zining Wang[*], Ruihao Gong[†], Jing Liu, Xinjie Zhang, Jinyang Guo, Xianglong Liu, Jun Zhang[†]
  ICML 2025.
- **TFMQ-DM: Temporal Feature Maintenance Quantization for Diffusion Models**
  Yushi Huang[*], Ruihao Gong[*], Jing Liu, Tianlong Chen, Xianglong Liu[†]
  CVPR 2024. ***Highlight.***
- **Temporal Feature Matters: A Framework for Diffusion Model Quantization**
  Yushi Huang, Ruihao Gong, Xianglong Liu[†], Jing Liu, Yuhang Li, Jiwen Lu, Dacheng Tao
  TPAMI 2025.
- **LLMC: Benchmarking Large Language Model Quantization with a Versatile Compression Toolkit**
  Ruihao Gong[*], Yang Yong[*], Shiqiao Gu[*], **Yushi Huang**[*], Chengtao Lv, Yunchen Zhang, Dacheng Tao, Xianglong Liu[†]
  EMNLP Industry 2024.
- **ProQuant: Progressive Quantization-aware Training for Edge MLLMs**
  Yufei Xue, **Yushi Huang**, Jiawei Shao[†], Pingcheng Dong, Yonghao Tan, Shiyao Li, Kwang-Ting Cheng, Jun Zhang[†]
  NeurIPS 2026.
- **Flash-VAED: Plug-and-Play VAE Decoders for Efficient Video Generation**
  Lunjie Zhu, **Yushi Huang**, Xingtong Ge, Yufei Xue, Zhening Liu, Yumeng Zhang, Zehong Lin, Jun Zhang[†]
  ICML 2026.
- **Light Forcing: Accelerating Autoregressive Video Diffusion via Sparse Attention**
  Chengtao Lv, Yumeng Shi, **Yushi Huang**, Ruihao Gong[†], Shen Ren, Wenya Wang[†]
  ICML 2026.
- **SALT: Self-Consistent Distribution Matching with Cache-Aware Training for Few-Step Video Generation**
  Xingtong Ge, Yi Zhang, **Yushi Huang**, Dailan He, Xiahong Wang, Bingqi Ma, Guanglu Song, Yu Liu, Jun Zhang[†]
  ECCV 2026.

Experience

Tencent · Hunyuan — Shenzhen, China

Research Intern; mentor: Yonglong Tian — *Oct. 2026 – Present*

Agentic RL for (M)LLMs.

Research Intern; mentor: Tianyu Pang — *Feb. 2026 – Oct. 2026*

RL for image and video generative models.

ByteDance · Seed — Shenzhen, China

Research Intern; mentor: Yang Zhang — *Sep. 2025 – Nov. 2025*

Inference acceleration for (M)LLMs.

Microsoft Research Asia — Beijing, China

Research Intern; mentor: Fangyun Wei — *Dec. 2024 – Feb. 2025*

Video generation and world models.

SenseTime Research — Beijing & Shenzhen, China

Research Intern (multiple terms); mentor: Ruihao Gong — *May 2023 – Feb. 2026*

Compression and acceleration for image and video diffusion models and LLMs.

Projects

LightCompress: Accurate and Efficient AIGC Model Compression — *700+ GitHub stars*

Core contributor:

- Implemented a broad suite of LLM quantization methods, including QuaRot, GPTQ, SmoothQuant, and OmniQuant.
- Built an end-to-end toolkit supporting multiple model architectures, evaluation protocols, and inference backends.
- Documented practical quantization recipes across model families and deployment settings.

LightX2V: Light Video Generation Inference Framework — *2.5k+ GitHub stars*

Contributor:

- Contributed to the framework design for adding Wan and HunyuanVideo model support.
- Integrated quantization, feature caching, and sparse attention features into the framework.
- Contributed to model distillation for LightX2V.

Academic Service

- **Conference reviewer**: NeurIPS, ICLR, ICML, COLM, AAAI, CVPR, and ECCV.

Skills

- **Programming & Frameworks**: Python, C, Java, PyTorch, and NumPy.
- **Research**: Reinforcement learning, model compression, and model acceleration.